

Chapter 6

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Chapter 6: Interdisciplinary STEM Projects - "STEM Innovator Challenge" Worksheet Hey, STEM Squad! This worksheet is your ticket to flexing your creativity and skills from Chapter 6. It's short, fun, and packed with challenges to test how Science, Tech, Engineering, and Math team up in your brain. Complete the tasks below, share ideas with friends, and let's see who's the ultimate STEM Innovator! Takes about 30-45 minutes. Grab a pen, some paper, and your imagination—let's roll!

Part 1: Quick Think - STEM Mashup (5 minutes) Task: Pick a teen problem (like slow school Wi-Fi or no hangout spots). In 3-4 sentences, brainstorm how you'd solve it using ALL four STEM fields. Example: For slow Wi-Fi, Science studies signal interference, Tech codes a signal tracker app, Engineering designs better router spots, and Math calculates cost per device. **Your Problem:** _____ **Your STEM Solution:**

Part 2: Mini-Design Challenge - Pick Your Project (15 minutes) Task: Choose ONE of the Chapter 6 projects below and sketch a quick idea or improvement. No perfection needed—just doodle and jot 2-3 notes on how STEM plays a role. Use the space or extra paper for your sketch.

- **Option A: Sustainable Mini-City** - Add a new "teen zone" (like a skate park). How does each STEM field help?
- **Option B: Water Filtration System** - Tweak the design for faster filtering. What STEM changes make it work?
- **Option C: Mobile App Concept** - Add a fun feature to solve a school issue. How does STEM shape it? **My Choice:** _____ **Sketch Space:** (Draw here or attach a separate sheet) **STEM Notes:**

-
-
-

Part 3: Pitch Power - Mini-Presentation Prep (10 minutes) Task: Imagine presenting your Part 2 idea to your class. Write a 2-3 sentence "hook" to grab attention at the start of your pitch. Then, list ONE cool visual or prop you'd show off. Example Hook: "Ever missed a school event 'cause no one told you? My app fixes that with instant alerts!" **My Hook:**

My Visual/Prop: _____ (e.g., a cardboard model, app sketch on phone)

Part 4: STEM Superhero Challenge - Fun Wrap-Up (5 minutes) Task: Turn yourself into a STEM Superhero based on this chapter! Pick a project or skill you vibed with most. Give yourself a hero name and tagline tied to your STEM power. Example: "Filter Flash - Purifying the Planet One Drop at a Time!" **My Hero Name:** _____ **My Tagline:** _____

Bonus Fun (Optional - 5 minutes) STEM Meme Moment: Create a quick meme caption or hashtag about combining STEM fields (like "#STEMSquadSavesTheDay" or "When Math Meets Magic!"). Write it below or share it in class for laughs! **My Meme Idea:** _____

Reflection for the Win: What's ONE thing from Chapter 6 you're most hyped to try IRL (like building a filter or

coding an app)? Why?